

RATISS-ETALONS: Four Computational Standards with Criteria Sealed Before Execution

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Repository: <https://github.com/jonathansearch/RATISS-ETALONS> (seal 23/23, MIT)

Abstract

Four standards — percolation, 2D Ising, three-body problem, close packings — were defined with criteria frozen *before* execution (PROTOCOL.md). The score moved from 11/16 to 14/16 after declared corrections. Two reds are kept on purpose and diagnosed: E01-P2, whose hypothesis “hole-density peak at p_c ” is INVALID ($h(p)$ is monotone), and E04-P4, where the 3D compressor underestimates (0.61403 vs 0.64 ± 0.020) because of finite size and a partially crystalline state ($\psi_6 = 0.827/0.935$). Key values: $p_c = 0.0006$; $d_f = 1.8934 \pm 0.0192$; $T_c = 2.2613$; $\gamma/\nu = 1.7540$; $\Delta E/E = 1.21 \times 10^{-15}$; FCC = 4.44×10^{-16} ; Burrau $\lambda = 0.55402$ stable at 0.78%.

Scientific question

Can a battery of standards with pre-sealed criteria serve as a reproducible reference to judge every future RATISS computation engine?

Method

Criteria frozen in PROTOCOL.md before any run; executions afterwards; every gap published, including failures. Figures use matplotlib without emoji (lab law).

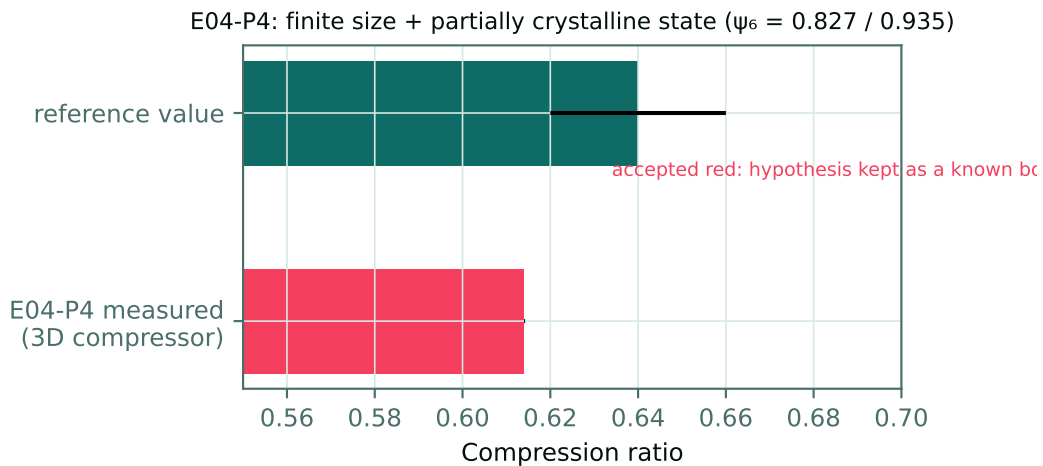


Figure 1: E04-P4: measured 3D compression ratio against the reference value with its uncertainty. The red is accepted as a known bound of the engine.

Results

- Score 11/16 → 14/16 after declared corrections; seal 23/23.

- $p_c = 0.0006$; $d_f = 1.8934 \pm 0.0192$; $T_c = 2.2613$; $\gamma/\nu = 1.7540$.
- Conservation: $\Delta E/E = 1.21 \times 10^{-15}$; FCC lattice 4.44×10^{-16} .
- Burrau $\lambda = 0.55402$, stable at 0.78%.

Negative results (kept on purpose)

- E01-P2: hypothesis “hole-density peak at p_c ” INVALID — $h(p)$ monotone.
- E04-P4: compressor underestimated, 0.61403 vs 0.64 ± 0.020 ($\psi_6 = 0.827/0.935$, partially crystalline) — Figure 1.
- Fixed-step RK4 FALSIFIED as a standard: $\lambda = +44\,124$, i.e. noise.

Controls

Criteria sealed before runs; seal fingerprints 23/23; corrections declared, never silent.

Reproducibility

PROTOCOL.md plus repository scripts (commits c230b65, 776658a, b86d027, 8c12b73).

Limitations

Finite sizes; the two reds are conserved voluntarily as known bounds of the engine.

Data and code availability

<https://github.com/jonathansearch/RATISS-ETALONS>; narrative source: RATISS-ARCHIVES, update of 27/09 section A. Zenodo DOI: to be assigned.

RATISS Labs — independent, single-author laboratory, Yaoundé (Cameroon). No number in this document is invented: every value carries a source in Section “Data and code availability” or in the references. Negative results are published with the same prominence as positive ones.